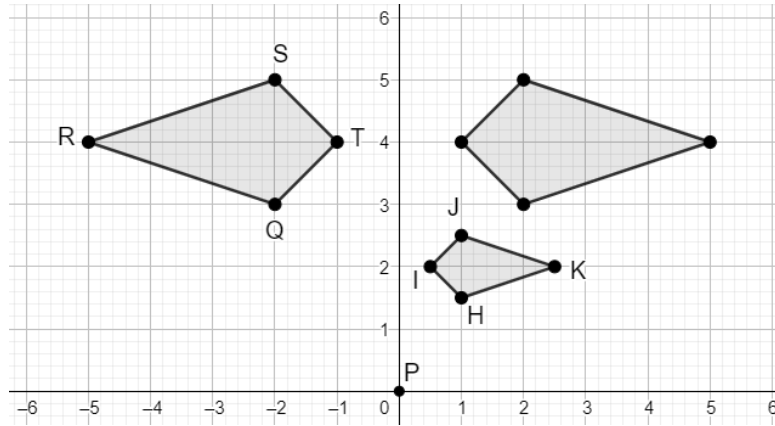
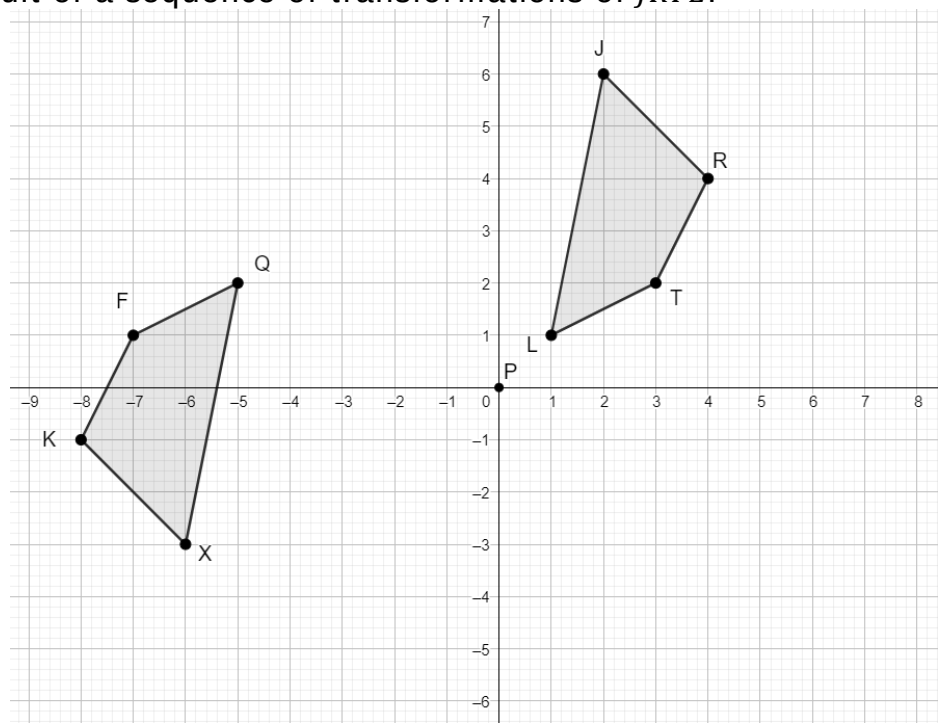


$HIJK$ is the result of a sequence of transformations of $QTSR$. The unlabelled quadrilateral is the first transformation.



1. Describe the first transformation of $QTSR$.
2. Write the transformation algebraically.
3. Describe the second transformation that results in $HIJK$.
4. Write the second transformation algebraically.
5. Explain if the sequence of transformations preserves distance.

XKFQ is the result of a sequence of transformations of *JRTL*.



Julia and Nathan were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Julia's Sequence	Written Description		Rotate 180° about the origin.
7.		Algebraic Description	$(x,y) \rightarrow (x + 4,y - 3)$	
8.	Nathan's Sequence	Written Description	Rotate 180° about the origin.	
9.		Algebraic Description		$(x,y) \rightarrow (x + 4,y - 3)$

10. Considering both proposed sequences, you conclude that correct.

- o Julia's sequence is
 - o Nathan's sequence is
 - o Both sequences are